

Angles, Quadrilaterals & Polygons

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Standard: CCSS.MATH.CONTENT.7.G.A.2 / 7.G.B.5 — Draw, describe, and analyze geometric figures; use facts about angles formed by parallel lines cut by a transversal; solve problems involving angle measures, quadrilateral properties, and polygon interior angle sums.



Mr. Kenny's Key Point

Cut it into triangles! Pick one vertex and draw all the diagonals — you get $(n - 2)$ triangles. Each triangle has 180° , so the polygon's interior angles add up to $(n - 2) \times 180^\circ$.

Or remember: all exterior angles ALWAYS add to 360° .

Part A — Parallel Lines Cut by a Transversal

Directions: Use the properties of parallel lines cut by a transversal to answer each question. Show all work and reasoning in the space provided.

- Two parallel lines are cut by a transversal, forming angles $\angle 1$ through $\angle 8$. Given that $m\angle 1 = 65^\circ$, find the measures of $\angle 2$, $\angle 5$, and $\angle 8$. Name the angle relationship you used for each.

Hint:

$\angle 1$ and $\angle 2$ are on a straight line (supplementary).

$\angle 1$ and $\angle 5$ are in the same position at each intersection.

$\angle 1$ and $\angle 8$ are on opposite sides of the transversal, outside the parallel lines.

$$m\angle 2 = \underline{\hspace{2cm}} \quad m\angle 5 = \underline{\hspace{2cm}} \quad m\angle 8 = \underline{\hspace{2cm}}$$

2. Angles $\angle 4$ and $\angle 6$ are formed where a transversal crosses two parallel lines.

(a) What type of angle pair are $\angle 4$ and $\angle 6$? _____

(b) Are they equal or supplementary? _____

(c) If $m\angle 4 = 112^\circ$, what is $m\angle 6$? _____

$$m\angle 6 = \underline{\hspace{2cm}}$$

3. Two parallel lines are cut by a transversal. Angles $\angle 3$ and $\angle 5$ are *alternate interior angles*, so they are equal.

Given $m\angle 3 = (2x + 10)^\circ$ and $m\angle 5 = (3x - 20)^\circ$, find the value of x and the measure of each angle.

$$x = \underline{\hspace{2cm}} \qquad m\angle 3 = m\angle 5 = \underline{\hspace{2cm}}$$

Directions: Use properties of quadrilaterals to solve each problem. Show all work in the space provided.

- $$m\angle B = \underline{\hspace{2cm}} \quad m\angle C = \underline{\hspace{2cm}} \quad m\angle D = \underline{\hspace{2cm}}$$

- What is $m\angle S$? Explain why these two angles are supplementary.

$$m\angle S = \underline{\hspace{2cm}}$$

- (a) Name this quadrilateral:

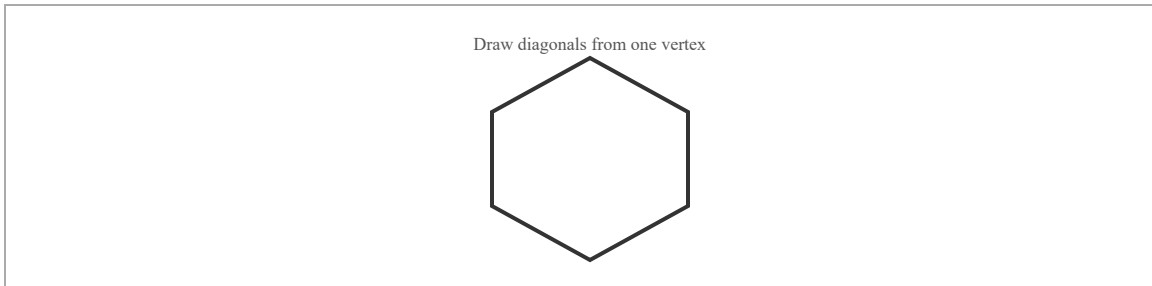
- Trapezoid Parallelogram Rectangle Rhombus Square

Part C — Polygon Angle Sum Theorem

Directions: Use $(n - 2) \times 180^\circ$ to find angle sums and individual angle measures. Show the formula and all steps.

7. Find the interior angle sum of a **hexagon** ($n = 6$) using the formula.

Then, in the hexagon below, *draw all the diagonals from one vertex* to show that a hexagon is made of $(n - 2) = 4$ triangles.



Interior angle sum = _____

8. Find the measure of **each interior angle** of a regular octagon ($n = 8$).

Hint: For a regular polygon, each interior angle = $\frac{(n - 2) \times 180^\circ}{n}$

Each interior angle = _____

9. The interior angle sum of a polygon is **1260°**.

(a) Use the formula $(n - 2) \times 180^\circ = 1260^\circ$ to find n .

(b) What is the name of this polygon?

$n =$ _____ Name: _____

10. Find the measure of each **exterior angle** of a regular pentagon ($n = 5$).

Hint: All exterior angles of any polygon always add up to 360°.

Bonus: Use your answer to find each *interior* angle of the regular pentagon. (Interior + Exterior = 180°)

Each exterior angle = _____ Each interior angle = _____